Dept. Computer Science and Engineering
University of Rajshahi

Semester Final Examination, B.Sc. Engg. 2019, 1st year, Odd semester

Course ID: CSE 1111 Course Title: Introduction to Computer Systems/Computer Fundamentals

Total Time 3 Hours Total Marks 52.5

(Answer any six questions taking three from each section)

Section A

1(a)	What do you mean by system boot up? Briefly discuss the PC's booting up process.	3
(b)	What is meant by computer generations? Explain different generations of computers.	3.75
(c)	Expand ABC, ENIAC, EDVAC and EDSAC.	2
2(a)	What is motherboard? Write the functions of it. Also mention the main components on a motherboard.	3
(b)	Briefly discuss how a keyboard works.	3
(c)	Explain the internal structure of CPU with diagram	2.75
3(a)	How does the control unit assist the CPU in carrying out its operations? Explain.	3
(b)	What do you understand by machine cycle? Explain the main operations those are accomplished during the machine cycle.	3
(c)	How does the bus width affect the overall speed of the computer system? Define the three types of I/O buses.	2.75
4(a)	Define printer. Explain the various criteria to evaluate a printer.	3
(b)	Describe the LCD monitor with its advantages.	3
(c)	Explain the four areas created on a magnetic disk during formatting.	2.75

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Section B

5(a)	What are difference between SRAM and DRAM? Why is a refreshing circuit needed for DRAM?	3
(b)	Write the characteristic of cache memory. What do you mean by L1, L2 and L3 cache?	2
(c)	Explain how data is stored on the surface of magnetic and optical disk	3.75
6(a)	Define the software. List and explain the types of software. Give two exaples of each category.	3
(b)	Explain three categories of operating systems with examples.	2.75
(c)	Define utility program. List a few names of utility programs with their tasks.	2
(d)	What is kernel?	1
7(a)	What is a computer network? Illustrate advantages of computer networks.	2.75
(b)	Explain the advantages and disadvantages of client-server and-peer-to-peer networks	3
(c)	Define network topology. Write down the advantages and disadvantages of bus, ring and star topologies.	3
8(a)	Define computer virus. How does computer virus work? Mention some notable computer viruses.	3
(b)	What is data processing? Define data and information.	2
(c)	Define Database Management System (DBMS). Explain the limitations of file based systems and write the advantages of DBMS.	3.75

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University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engineering Part-1 Odd Semester Examination 2018
Course Code: **CSE1111** Course Title: **Introduction to Computer Systems**
Marks: 52.50 Time: 3 Hours

(Answer Six (06) Questions taking any Three (03) Questions from each Section)

Section-A

1.(a)	What is Computer? Explain the characteristics of Computer.	3
(b)	What are the basic operations of Computer? Describe them.	3.75
(c)	Compare microcomputers, minicomputers and main frames in terms of size and cost.	2
2. (a)	What are the different forms to represent negative signed binary numbers? Explain them with example.	3
(b)	Evaluate the binary number 10001111 in base 10 under (i) Signed magnitude notation and (ii) Two's complement notation.	3
(c)	Convert the followings (i) $(510.125)_{10}=()_2$? (ii) $(AEF5)_{16}$ to binary and octal.	2
(d)	Perform the subtraction 11110111-10110111 using 2's complement method.	0.75
3. (a)	Briefly describe the PC's system booting-up sequence.	3
(b)	Describe the function of the CPU as fetching and executing instructions stored in memory.	3
(c)	Explain the different factors affecting the processing speed of CPU.	2.75
4. (a)	Differentiate between LCD and LED monitors.	2.75
(b)	Discuss the printing mechanism of a laser printer.	3
(c)	Define the following terms: (i) Optical Mouse, (ii) Bar-code reader, and (iii) Modem.	3

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Section-B

5. (a)	What is Memory? Draw the memory hierarchy according to their access speed.	2
(b)	What are the features of storage devices? Describe them.	3
(c)	Explain how data is stored on the surface of magnetic disk.	2
(d)	A 6 platter hard disk has 600 tracks per surface. There are 10 sectors per track and 512 bytes per sector. What is the storage capacity of the disk? How many cylinders does the disk pack have?	1.75
6. (a)	Define Operating System (OS) and explain functions of OS.	2.5
(b)	Explain three categories of OS with examples.	3
(c)	Differentiate between DOS and Windows.	2
(d)	Why is virtual memory needed?	1.25
7. (a)	Define computer networks. Mention four benefits that computer networks provide to their users.	2.75
(b)	Explain network topology. Discuss the function of Star and Bus topology and differentiate between them.	4
(c)	Define the following terms (i) IP address (ii) TCP/IP, (iii) SMTP, (iv) http.	2
8. (a)	Mention three utility programs with its purposes.	3
(b)	Define computer virus. Explain the characteristics of it.	2.5
(c)	How do viruses infect and spread in a computer system?	2
(d)	Mention different types of viruses with examples.	1.25

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University of Rajshahi
(For Affiliated Colleges)
Department of Computer Science and Engineering
B.Sc. Engineering Part-1 Odd Semester Examination 2021
Course: CSE1111 (Introduction to Computer Systems)
(According to the Syllabus 2019-2020)
Marks: 52.50 Time: 3 Hours
(Answer any THREE questions from each Section)

Section-A

1.(a)	What is Computer? Briefly explain different types of computer according to their size.	4
(b)	What is computer generation? Explain the features of 1st generation and 3rd generation computers.	2.75
(c)	Explain the basic operations of computer?	2
2. (a)	Write the functions of motherboard. Also mention the main components on a motherboard.	3
(b)	Draw the block diagram of a digital computer and explain the functions of ALU and CU.	4
(c)	Why computer is known as a data processing system?	1.75
3. (a)	Draw the architecture of a 64x4 RAM.	2.5
(b)	Suppose your computer have 2 MB cache, 8 GB RAM and 1 TB hard disk drive. Why three different types of memory have been used- Explain.	2.5
(c)	A hard disk has 12 plates with 80 tracks of each surface. Each track is divided into 20 sectors and each sector contains 1024 KB of data. Calculate the capacity of the hard disk.	3.75
4. (a)	What is printer? Explain the evaluation criteria of printers.	4
(b)	Differentiate between impact printer and non-impact printer.	2
(c)	Distinguish between Active Matrix LCD and Passive Matrix LCD monitor.	2.75

Section-B

5. (a)	Define system software and application software with examples.	3
(b)	Distinguish between DOS and Windows operating systems.	4
(c)	What are the benefits of using LINUX over Windows operating system?	1.75
6. (a)	Define computer network. What do you mean by Hub, Bridge and Switch?	4
(b)	Define network topology. Explain the bus topology with its merits and demerits.	3
(c)	Explain UTP and Co-axial cables used in communication.	1.75
7. (a)	What do you mean by DBMS? Mention some advantages of DBMS.	3
(b)	Distinguish between data and information.	2.5
(c)	Differentiate between flat file databases and relational databases.	3.25
8. (a)	Define computer virus. What are the effects of computer virus in computer?	3
(b)	How computer virus can be removed? Explain.	2.75
(c)	What do you mean by data redundancy and data inconsistency?	3

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University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engineering Part-1 Odd Semester Examination 2021
Course: CSE1111 (Introduction to Computer Systems)
Full Marks: 52.50 **Time: 3 Hours**

[Answer Six questions taking at least Three from each section]

Section-A

1.(a)	Define computer. Explain the evolution of computer.	3
(b)	What do you mean by computer generation? Specify the electronic components and languages used in different generation. Give one example of each generation's computer.	3.75
(c)	What is an IC? How does it reduce the size of computer?	2
2. (a)	What is meant by 3MHz CPU? How does the increasing number of cores will enhance the performance of CPU?	3
(b)	State the three steps that CPU continuously carries out in a cycle when processing a program and explain what happens at each step.	2.75
(c)	Write the functions carried out by address bus, data bus and control bus when the CPU uses to perform certain operation.	3
3. (a)	Briefly discuss the data read-write mechanism of a hard disk drive.	4
(b)	A hard disk has 10 plates with 80 tracks of each surface. Each track is divided into 20 sectors and each sector contains 512KB of data. Calculate the capacity of the hard disk in GB.	1.75
(c)	Explain the working principle of an optical disk drive.	3
4. (a)	List three types of printers that are regularly used. Write the characteristics of impact and nonimpact printers.	3
(b)	Describe how a dot matrix printer creates an image on a page.	2.75
(c)	Write the general use of an MICR. What are OCR and OMR?	3

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Section-B

5. (a)	Differentiate between the characteristics of primary and secondary storage of a computer system.	3
(b)	What is SSD? Explain the advantages and disadvantages of it.	2
(c)	In the context of magnetic disk storage, define (i) Access time, (ii) Seek time (iii) Latency, and (iv) Transfer rate.	2
(d)	How does cache memory reduce the speed mismatch between processor and main memory?	1.75
6. (a)	Define system software and application software with example.	2
(b)	What is an operating system? Mention the types and functions of operating systems.	3.75
(c)	What is utility program? List some utility programs with their functions.	3
7. (a)	What is computer network? Discuss the advantages and risks of computer network.	3
(b)	Explain different types of computer network.	2.75
(c)	Define guided media. Explain different guided media used for data communication.	3
8. (a)	What are computer virus, trojan and worm? Write possible symptoms of a virus affected computer.	3
(b)	Write down the steps to protect your computer from virus attack.	2.75
(c)	Discuss the main characteristics of the database approach and how it differs from traditional file processing approach.	3

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University of Rajshahi
Dept. of Computer Science and Engineering
B.Sc. (Engg.), Part-I Odd Semester, Examination-2017
Course: CSE-1111 (Computer Fundamentals)

Full Marks-52.5 **Time: 3 hours**

[Answer any **SIX (06)** questions taking **THREE (03)** from each section]

Section-A

1. a) Define computer. Discuss different types of digital computers with their applications. 5
   b) Draw the block diagram of a digital computer and discuss how a program is executed by it. 3.75

2. a) What is CPU? Draw the block diagram of the internal structure of CPU and write its functions. 4
   b) Draw the architecture of a 64×4 RAM and discuss data read-write mechanism in it. 3
   c) Write differences between SRAM and DRAM. 1.75

3. a) Draw the architecture of a 64×4 RAM and discuss how data is read from and write into it. 4
   b) Briefly discuss the data read-write mechanism of a hard disk drive.
   c) A hard disk has 10 plates with 80 tracks of each surface. Each track is divided into 20 sectors and each sector contains 512KB of data. Calculate the capacity of the hard disk. 3 1.75

4. a) Compare CRT, LCD and LED monitors. 3
   b) Write the differences between impact and non-impact printers. 2
   c) What do you mean by system buses? Explain. 2
   d) What are the functions of BIOS? 1.75

Section-B

5. a) What do you mean by data and information? How does computer process data? 3
   b) What do you mean by database and database management system? Are they same or different? How? 2.75
   c) What is MODEM? Why it is used? 3

6. a) What is utility program? Mention some utility programs with their functions. 3
   b) What is language translator? Briefly discuss different types of language translator programs. 2.75
   c) Define the following types of software: 3
      (i) Driver software (ii) User-written program (iii) Package program.

7. a) What is operating system? What are the characteristics of the operating system? 3.5
   b) Define computer virus. Differentiate virus with malware, spyware and spam. 3.5
   c) Define anti-virus and computer hacker. 1.75

8. a) What is computer network? Write the major purposes to create a computer network. 2.75
   b) Define network topology and network protocol with examples. 2
   c) What is internet? Define the following terms related to internet: 4
      (i) Browser (ii) Search engine (iii) WWW.

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University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engineering Part-1 Odd Semester Examination 2018
Course Code: **CSE1111** Course Title: **Computer Fundamentals**
Full Marks: 52.50 Time: 3 Hours
(According to syllabus 2016-2017)

[Answer Six (06) Questions taking any Three (03) Questions from each Section]

Section-A

1.(a)	Define computer? Discuss different types of computers with their applications.	3
(b)	Explain computer generation. Mention the development technologies used in different generation of computers.	2.75
(c)	Draw the block diagram of a digital computer system and discuss the function of the different units of a computer system.	3
2. (a)	What are the different forms to represent negative signed binary numbers? Explain them with example.	3
(b)	Evaluate the binary number 10001111 in base 10 under (i) Signed magnitude notation and (ii) Two's complement notation.	3
(c)	Convert the followings (i) $(510.125)_{10}=()_2$? (ii) $(AEF5)_{16}$ to binary and octal.	2
(d)	Perform the subtraction 11110111-10110111 using 2's complement method.	0.75
3. (a)	Briefly describe the PC's system booting-up sequence.	3
(b)	Draw the block diagram of a digital computer and discuss how a program is executed by it.	4
(c)	Explain the functions of CPU.	1.75
4. (a)	What is printer? Explain about the various criteria for evaluating a printer.	3
(b)	Explain impact printer and non-impact printer.	2
(c)	Describe the working principle of a Laser printer.	2.75
(d)	How does the bus width affect the overall speed of the computer system?	1

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Section-B

5. (a)	What is Memory? Draw the memory hierarchy according to their access speed.	2
(b)	Explain how data is transferred from an optical disc (i.e. CD-ROM) to the main memory.	5
(c)	A 6 platter hard disk has 600 tracks per surface. There are 10 sectors per track and 512 bytes per sector. What is the storage capacity of the disk? How many cylinders does the disk pack have?	1.75
6. (a)	What is software? Define system software and application software with examples.	3
(b)	Discuss operating system with its types and functions.	3
(c)	Explain package program and user-written program with example.	2.75
7. (a)	What is computer network? Compare among LAN, MAN and WAN.	3
(b)	Define network topology. Discuss bus topology used in a computer network.	3
(c)	How does a request from a computer sent over the Internet? Describe.	2.75
8. (a)	Mention three utility programs with its purposes.	3
(b)	Define computer virus. Explain the characteristics of it.	2.5
(c)	How does virus infect and spread in a computer system?	2
(d)	Mention different types of viruses with examples.	1.25

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University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engineering **Part-I Odd** Semester Examination **2020**
Course Code: **CSE1111** Course Title: **Introduction to Computer Systems**
Time: **03** Hours Full Marks: **52.5**

[N.B. Answer 06(Six) questions taking any 03(Three) questions from each section. The figures in the right margin indicate full marks.]

Section-A

1.(a)	Define computer. Explain the characteristics of computer.	3
(b)	What is meant by computer generations? Why were the first generation computers huge in size? State the features of the third generation computers.	3.75
(c)	Distinguish between mainframe and supercomputer.	2
2. (a)	What is CPU? Draw the internal block diagram of a CPU.	3.75
(b)	Explain the importance of a bus in the computer system. What are the different types of buses found in the computer system?	3
(c)	Why cache memory is needed in a computer system? Explain it with block diagram.	2
3. (a)	What is chipset? Explain the main activities of the chipset on a motherboard.	3
(b)	Define computer port. Give the two examples of each of serial port and parallel port.	2
(c)	What is core processor? Explain core i5 processor. Which type of RAM is used in core i5 processor?	3.75
4. (a)	Briefly explain the types of data storage and its relation with CPU.	3
(b)	Explain the principle to store data in magnetic and optical storage devices.	3
(c)	Give a complete specification to buy a modern computer.	2.75

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Section-B

5. (a)	Classify various types of Memory on the basis of the material used.	3
(b)	Explain Memory Hierarchy according to access time required for memory.	2
(c)	How many address and data lines will be there for a 16M x 32 memory system?	1.75
(d)	Explain about SSD. Discuss advantages of it.	2
6. (a)	Explain different types of OS and also explain different types of tasks done by OS.	3
(b)	Write three utility programs with its purposes.	2.75
(c)	Explain the DOS internal structure.	3
7. (a)	What is computer network? Give the comparison of different types of computer network.	3.5
(b)	Define topology. Explain the factors that affect choice of topology for a network.	2.25
(c)	Explain the function of following network devices (i) Router, (ii) Switch, (iii) Gateway.	3
8. (a)	Define Malware. Explain how does it spread in computer and how does protect from it.	2
(b)	What is data processing? Explain different stages of data processing cycle.	3
(c)	What is data redundancy and data inconsistency? How DBMS is superior to conventional file processing system?	3.75

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University of Rajshahi
(Affiliated College)
Department of Computer Science and Engineering
B.Sc. Engineering Part-1 Odd Semester Examination 2021
Course: CSE1111 (Introduction to Computer Systems)
Marks: 60 Time: 3 Hours

(Answer any THREE questions from each Section)

Section-A

1.(a)	What is Computer? Classify of the Computer and explain them.	2
(b)	Discuss the computer generation focus with key hardware, software technologies and key characteristics.	4
(c)	Draw the block diagram of a digital computer and discuss how a program is executed by it.	4
2. (a)	Briefly describe the PC's system booting-up process step by step.	2
(b)	Draw the architecture of a 64x4 RAM and discuss how data is read from and write into it.	6
(c)	Explain how does the keyboard work?	2
3. (a)	What is Cathode Ray Tube (CRT)? Explain the working principle of a CRT monitor with proper diagram.	4
(b)	Differentiate between impact printer and non-impact printer.	2
(c)	What is clock speed? How do you choose between more processor cores or a higher clock speed?	4
4. (a)	What is printer? Describe the working principle of a Laser printer.	3
(b)	Explain the two phases for executing an instruction in a computer with appropriate example.	4
(c)	Define computer bus and discuss different types of computer buses.	3

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Section-B

5. (a)	What is memory? Draw the memory hierarchy according to their access speed.	3
(b)	Differentiate between primary memory and secondary memory.	2
(c)	Explain the role of cache memory in a computer system.	3
(d)	What are differences between SRAM and DRAM?	2
6. (a)	What is software? Define Operating System (OS) with its functions and types.	4
(b)	What is virtual memory? Discuss the implementation of virtual memory.	4
(c)	A hard disk has 12 plates with 80 tracks of each surface. Each track is divided into 20 sectors and each sector contains 512 KB of data. Calculate the capacity of the hard disk.	2
7. (a)	What is computer network? Classify the computer network based on size (geographic area covered) and explain them.	4
(b)	Define network topology. Explain the two different ways by which you can connect to the network.	3
(c)	What is communication media in networking? Mention various communication media with examples	3
8. (a)	How is DBMS superior to conventional file processing? Explain.	3
(b)	Define computer virus. Distinguish among computer virus, worm, and trojan horse.	4
(c)	Define computer hacker. Discuss some common hacking methods.	3

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